Lab Practice Work: Sorting Algorithms

In this Practice, you will use the Basic World Cities Database. It can be downloaded for free from https://simplemaps.com/data/world-cities

Problem 1. BubbleSort

At first, use the unique latitude values of each city only.

1. Implement a proper bubble sort algorithm so that all city latitudes are in an ordered list.
2. Count the number of merges needed to sort the dataset. Does it change if you randomly order the list before sorting? Why/why not?

Problem 2. MergeSort

At first, use the unique latitude values of each city only.

1. Implement a proper merge sort algorithm so that all city latitudes are in an ordered list.
2. Count the number of merges needed to sort the dataset. Does it change if you randomly order the list before sorting? Why/why not?

Problem 3: QuickSort

At first, use the unique latitude values of each city only.

1. Implement a proper quick sort algorithm so that all city latitudes are in an ordered list.
2. Count the number of comparisons needed to sort the dataset. Does it change if you randomly order the list before sorting? Why/why not?

Problem 4: InsertionSort

At first, use the unique latitude values of each city only.

1. Implement a proper insertion sort algorithm so that all city latitudes are in an ordered list.
2. Count the number of comparisons needed to sort the dataset. Does it change if you randomly order the list before sorting? Why/why not?